

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (IT)

Dec 2013

IT706 Information Security, Security policies & Assurance

Date: 26.12.2013, Wednesday

Time: 10:00 am To 01:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) What is confidentiality? How it differs from integrity? [03]
- (b) Explain security service and security mechanism. Relate security service and security mechanism. [04]
- Q - 2 (a) What is cryptanalysis? Explain four types of cryptanalysis attacks. [05]
- (b) Explain mono alphabetic and poly alphabetic cipher. Which is more secure? Why? [05]
- (c) Apply a brute-force attack to break the cipher "UVACLYFZLJBYL". Assume that algorithm used to encipher the message is Caesar cipher. [04]

OR

- Q - 2 (a) Generate play fair cipher key matrix with key "Cryptography". Find the cipher text of "Hello". [05]
- (b) The encryption key in a transposition cipher is (3, 2, 6, 1, 5, and 4). Find the decryption key. [05]
- (c) What is substitution cipher? Write the differences between rotor cipher and enigma cipher? [04]
- Q - 3 (a) List and explain the components of modern block ciphers. Explain diffusion and confusion. [07]
- (b) Explain the Vigenere cipher with key with example. [07]

OR

- Q - 3 (a) Write the differences between p-box and s-box. What is diffusion and confusion in a block cipher? [07]
- (b) Draw the block diagram of the DES function. How many permutations are used in a DES cipher algorithm and round key generator? Why does the DES function need expansion? [07]

SECTION – II

- Q - 4 Explain the Structure of each round at the encryption site of AES cipher. [07]
- Q - 5 (a) What is CBC? What is Cipher text Stealing? Explain with example. [07]
- (b) Decrypt the following cipher text using RSA algorithm (# is used as delimiter): [07]

006 # 507 # 158 # 524 # 065 # 020 # 158 # 012

N=533, E=353

OR

- Q - 5 (a) Define Kerberos and name its server. Briefly explain the duties of each server. [07]
- (b) In the Diffie-Hellman protocol, $g=7$, $p=23$, $x=3$ and $y=5$. [07]
- a. What is the value of the symmetric key?
- b. What is the value of R1 and R2?
- Q - 6 Write a short note on any *TWO*. [14]

1. X.509 digital certificate
2. Master key generation in SSL
3. PGP
